

Lab Report: EEE 4878

Design and Characterization of a Bandgap Reference Voltage Circuit

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ABSTRACT

A bandgap reference circuit is an essential analog integrated circuit block used to generate a stable voltage reference that is independent of temperature and supply voltage variations. In this experiment, a CMOS-based bandgap reference voltage generator was designed and simulated to study its operating principle and performance characteristics.

The circuit was implemented using Cadence Virtuoso with the GPDK090 technology library. An ideal operational amplifier was created using a voltage-controlled voltage source, and the complete bandgap reference circuit was analyzed through DC simulation, temperature sweep, and supply voltage variation analysis. The temperature compensation technique was investigated by combining the negative temperature coefficient of the base-emitter voltage with the positive temperature coefficient of the thermal voltage component.

The simulation results showed that the designed circuit generated a reference voltage of $0.8136V$ at a supply voltage of $3.3V$, which closely matched the expected value of $0.814V$. The variation of the reference voltage was observed over a temperature range of $10^{\circ}C$ to $50^{\circ}C$, and the temperature coefficient at $27^{\circ}C$ was calculated to be $-1.2066mV/^{\circ}C$. The supply voltage sweep from $3.1V$ to $3.5V$ also demonstrated stable reference voltage performance.

The experiment verified the theoretical operation of a bandgap reference circuit and demonstrated its importance in precision analog applications such as voltage regulators, ADCs, DACs, and power management integrated circuits.

KEYWORDS

1. Bandgap
2. Reference
3. CMOS
4. Voltage
5. Temperature

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Chapter 1

INTRODUCTION

A bandgap reference circuit is an important analog integrated circuit block used to generate a stable reference voltage that remains nearly constant with variations in temperature, supply voltage, and process parameters. In this experiment, a CMOS-based bandgap reference voltage generator is designed and simulated to study its operating characteristics.

The main objective of this experiment is to understand the operating principle of bandgap reference circuits and learn how temperature-dependent voltages can be combined to achieve a stable reference voltage. The experiment also provides practical experience in designing and simulating analog circuits using CMOS technology.

During this laboratory experiment, the performance of the designed bandgap reference circuit was analyzed through simulation. The reference voltage variation with temperature and supply voltage was investigated, and the temperature coefficient of the circuit was evaluated to determine its stability.

Bandgap reference circuits are widely used in real-world electronic systems such as analog-to-digital converters (ADC), digital-to-analog converters (DAC), voltage regulators, and power management ICs. A stable reference voltage improves the accuracy and reliability of these systems, making bandgap circuits essential components in modern integrated circuits.

Chapter 2

THEORY

A bandgap reference circuit generates a temperature-independent voltage by combining two voltages with opposite temperature coefficients. The base-emitter voltage (V_{BE}) of a bipolar transistor decreases with increasing temperature, providing a negative temperature coefficient, while the difference in base-emitter voltage (ΔV_{BE}) between two bipolar transistors increases with temperature, providing a positive temperature coefficient.

The reference voltage is obtained by adding the scaled thermal voltage component to the base-emitter voltage. The output reference voltage can be expressed as:

$$V_{REF} = |V_{BE}| + \frac{R_2}{R_1} V_T \ln(n) \quad (2.1)$$

where V_T is the thermal voltage, R_1 and R_2 are resistor values, and n represents the ratio of transistor emitter areas or current densities.

By properly selecting the resistor ratio, the positive and negative temperature coefficients can be balanced, resulting in a reference voltage with a very small temperature dependence. This temperature compensation property makes bandgap reference circuits suitable for precision analog applications.

2.1 Pre-lab Derivation

2.1.1 Derivation of Equation (1)

The output voltage of the bandgap reference circuit is generated by combining a negative temperature coefficient voltage (V_{BE}) with a positive temperature coefficient voltage (ΔV_{BE}).

For two bipolar transistors operating at different current densities, the difference in their base-emitter voltages is given by:

$$\Delta V_{BE} = V_{BE1} - V_{BE2} \quad (2.2)$$

The base-emitter voltage difference can also be expressed using the thermal voltage equation:

$$\Delta V_{BE} = V_T \ln(n) \quad (2.3)$$

where,

$$V_T = \frac{kT}{q} \quad (2.4)$$

is the thermal voltage, k is Boltzmann's constant, T is absolute temperature, and q is the electron charge. The factor n represents the ratio of emitter areas or current densities of the bipolar transistors.

The voltage developed across resistor R_1 is:

$$V_{R1} = \Delta V_{BE} = V_T \ln(n) \quad (2.5)$$

The current flowing through R_1 is therefore:

$$I_{R1} = \frac{V_T \ln(n)}{R_1} \quad (2.6)$$

The same current flows through resistor R_2 , producing an additional voltage:

$$V_{R2} = I_{R1} R_2 \quad (2.7)$$

Substituting the value of current:

$$V_{R2} = \frac{R_2}{R_1} V_T \ln(n) \quad (2.8)$$

The output reference voltage is the summation of the base-emitter voltage and the amplified thermal voltage component:

$$V_{REF} = |V_{BE3}| + V_{R2} \quad (2.9)$$

Therefore,

$$V_{REF} = |V_{BE3}| + \frac{R_2}{R_1} V_T \ln(n) \quad (2.10)$$

which is the required Equation (1).

2.1.2 Derivation of Equation (2)

The temperature variation of the reference voltage can be obtained by differentiating Equation (1) with respect to temperature.

Starting from:

$$V_{REF} = |V_{BE3}| + \frac{R_2}{R_1} V_T \ln(n) \quad (2.11)$$

Taking the derivative:

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE}}{\partial T} + \frac{R_2}{R_1} \ln(n) \frac{\partial V_T}{\partial T} \quad (2.12)$$

Since,

$$V_T = \frac{kT}{q} \quad (2.13)$$

therefore,

$$\frac{\partial V_T}{\partial T} = \frac{V_T}{T} \quad (2.14)$$

Substituting this value:

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{BE}}{\partial T} + \frac{V_T}{T} \ln(n) \frac{R_2}{R_1} \quad (2.15)$$

The temperature dependence of the base-emitter voltage is given by:

$$\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4 + m)V_T - E_g/q}{T} \quad (2.16)$$

where E_g is the silicon bandgap energy and m is the temperature exponent of carrier mobility.

Therefore, the temperature coefficient of the bandgap reference becomes:

$$\frac{\partial V_{REF}}{\partial T} = \frac{V_{BE} - (4 + m)V_T - E_g/q}{T} + \frac{V_T}{T} \ln(n) \frac{R_2}{R_1} \quad (2.17)$$

This is the required Equation (2). The first term has a negative temperature coefficient, while the second term has a positive temperature coefficient. By properly selecting the resistor ratio R_2/R_1 , both temperature effects can cancel each other, resulting in a stable reference voltage.

Chapter 3

LAB HANDOUT QUESTIONS

Question 1

Simulate the circuit with $V_{DC} = 3.3V$ and find the value of V_{REF} . The expected value of V_{REF} is 0.814V.

Question 2

Show the variation of the reference voltage with temperature (10°C - 50°C).

Question 3

What is the temperature coefficient (TC) of the reference voltage at room temperature (27°C)?

Question 4

Show the variation of the reference voltage with variation of supply voltage (3.1V - 3.5V).

Question 5

Can the temperature coefficient (TC) be made zero at room temperature? When $TC = 0$, the reference voltage (V_{REF}) versus temperature (T) curve will exhibit a maximum or minimum, i.e., it will be horizontal at a specific temperature.

Chapter 4

TOOLS USED

- Cadence Virtuoso
- Spectre Simulator
- GPDK090 Technology Library
- MATLAB / Python (if applicable)

Chapter 5

PROCEDURE

The bandgap reference voltage generator circuit was designed and simulated using Cadence Virtuoso with the GPDK090 technology library. The complete procedure was carried out in the following steps:

1. Circuit Schematic Creation

The schematic of the bandgap reference circuit was created according to the given circuit topology. An ideal operational amplifier was implemented using a voltage-controlled voltage source (VCVS) from the analogLib library. The gain of the VCVS was set to 100000 to represent an ideal op-amp behavior.

2. Circuit Block and Symbol Generation

A separate circuit block was created for the ideal op-amp. Input and output pins were added, and the symbol view was generated using the Create Cellview option. The generated symbol was then used in the main bandgap reference schematic.

3. Parameter Configuration

The required transistor and resistor parameters were configured. The bipolar transistors were selected from the GPDK090 library, and the required multiplication factors and device dimensions were assigned. The resistor values were set according to the design requirements.

4. Simulation Setup

The GPDK090 model files were added through the ADE model library setup. A DC simulation was configured with a supply voltage of 3.3V. Temperature sweep and supply voltage sweep simulations were also configured to analyze circuit performance.

5. Data Collection

The output reference voltage (V_{REF}) was measured from the simulation results. The variation of reference voltage with temperature and supply voltage was recorded using simulation plots.

6. Analysis of Simulation Results

The obtained simulation results were analyzed to determine the reference voltage accuracy, temperature coefficient, and supply voltage sensitivity of the designed bandgap reference circuit.

Chapter 6

RESULTS

The designed bandgap reference voltage generator circuit was simulated using Cadence Virtuoso. The simulation results were analyzed to determine the reference voltage value, temperature dependency, temperature coefficient, and supply voltage sensitivity.

6.1 Task 1: DC Simulation of Reference Voltage

Objective:

Simulate the bandgap reference circuit with a supply voltage of $V_{DC} = 3.3V$ and determine the output reference voltage V_{REF} .

Circuit Schematic:

The complete bandgap reference circuit schematic used for simulation is shown below.

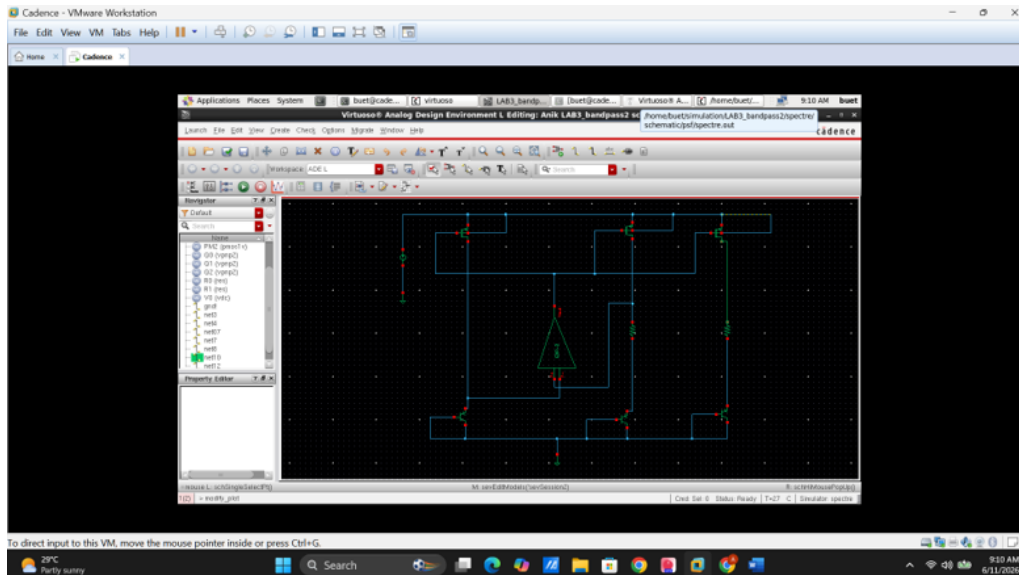


Figure 6.1: Schematic of the bandgap reference voltage generator circuit

Simulation Result:

The DC simulation was performed with a supply voltage of 3.3V. The obtained reference voltage from the simulation is shown below.

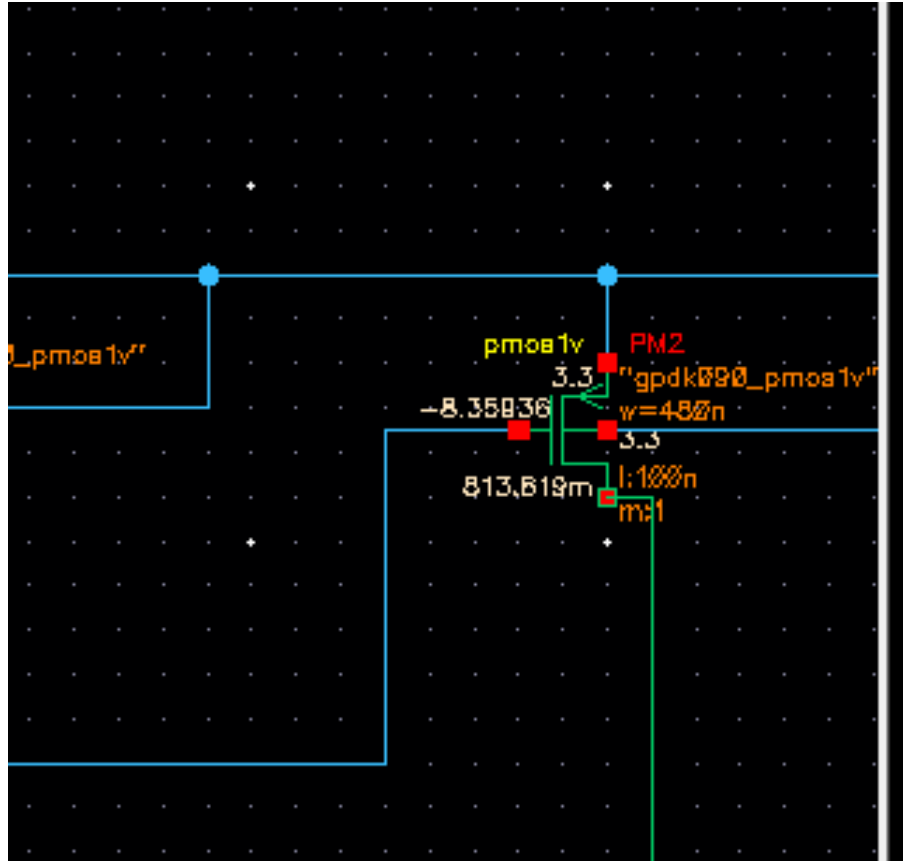


Figure 6.2: DC simulation result showing reference voltage

The measured output reference voltage was:

$$V_{REF} = 0.8136V \quad (6.1)$$

The obtained value is compared with the expected value of 0.814V.

6.2 Task 2: Temperature Variation of Reference Voltage

Objective:

Analyze the variation of reference voltage with temperature from 10°C to 50°C.

A temperature sweep simulation was performed to observe the change in V_{REF} with temperature.

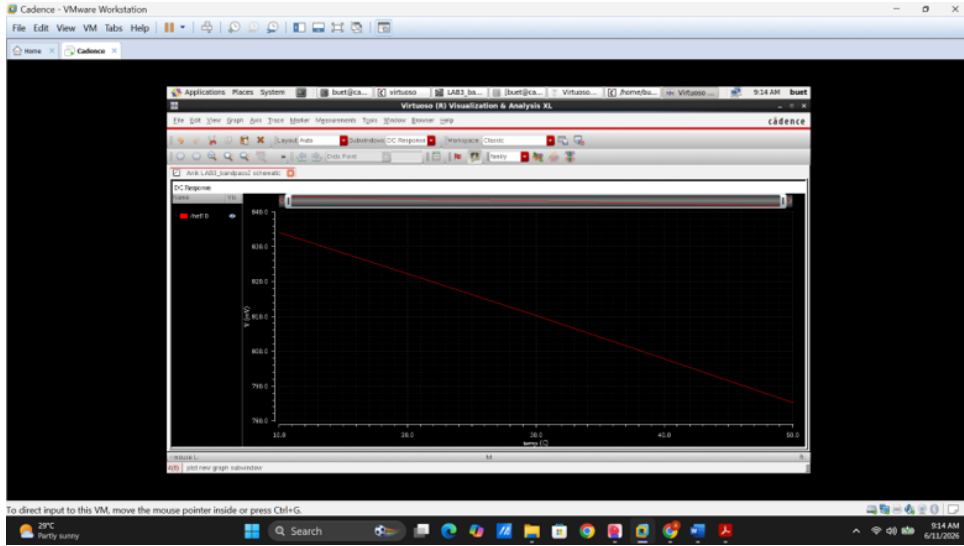


Figure 6.3: Variation of reference voltage with temperature

The graph shows the temperature dependence of the reference voltage. The small variation in output voltage indicates the temperature compensation ability of the bandgap reference circuit.

6.3 Task 3: Temperature Coefficient Calculation

Objective:

Calculate the temperature coefficient (TC) of the reference voltage at room temperature (27°C).

The temperature coefficient is calculated using:

$$TC = \frac{\Delta V_{REF}}{\Delta T} \quad (6.2)$$

At 27°C, the calculated temperature coefficient is:

$$TC = -1.2066mV/^{\circ}C \quad (6.3)$$

A lower temperature coefficient indicates better stability of the reference voltage against temperature variations.

6.4 Task 4: Supply Voltage Variation

Objective:

Observe the variation of the reference voltage with supply voltage variation from 3.1V to 3.5V.

The supply voltage sweep simulation was performed and the output variation of V_{REF} was recorded.

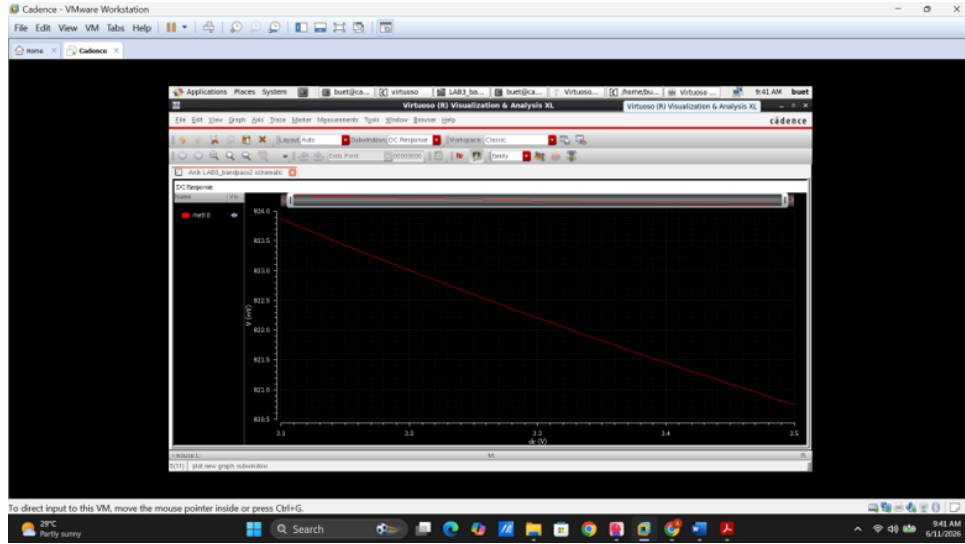


Figure 6.4: Variation of reference voltage with supply voltage

The result demonstrates the supply sensitivity of the bandgap reference circuit.

6.5 Task 5: Possibility of Zero Temperature Coefficient

Objective:

Determine whether the temperature coefficient can be made zero at room temperature.

The temperature coefficient can be adjusted close to zero by selecting an appropriate resistor ratio $\frac{R_2}{R_1}$. When:

$$\frac{\partial V_{REF}}{\partial T} = 0 \quad (6.4)$$

the positive temperature coefficient of ΔV_{BE} compensates for the negative temperature coefficient of V_{BE} .

At this condition, the V_{REF} versus temperature curve becomes horizontal at a particular temperature, representing a maximum or minimum point of the curve depending on the circuit characteristics.

6.6 Task: Creation of 1 V bandgap reference generator circuit

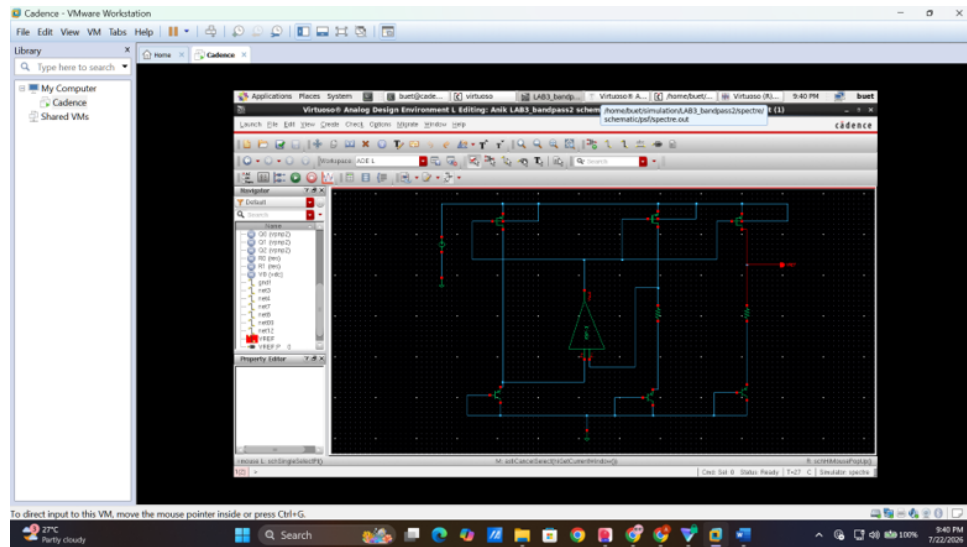
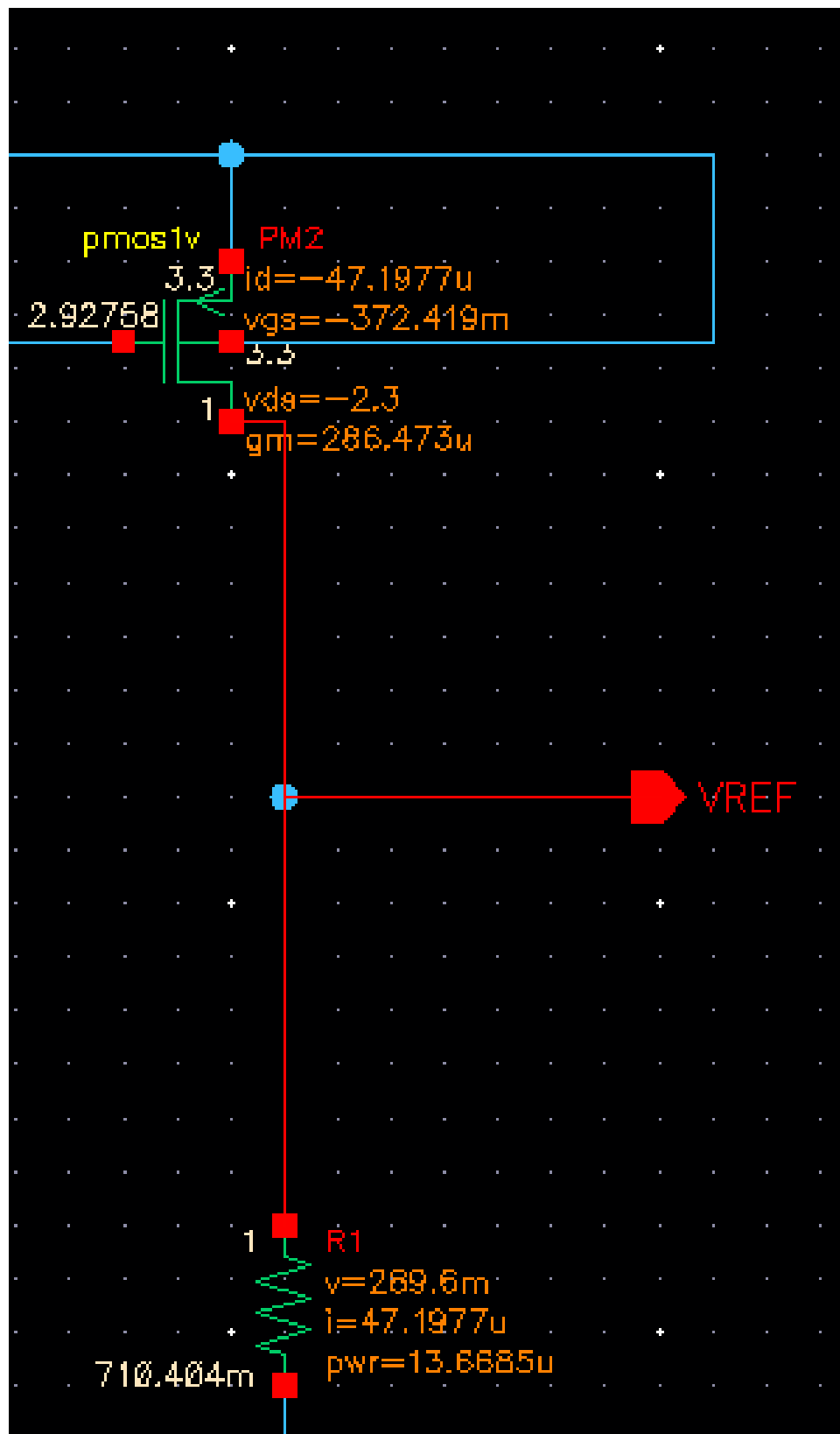


Figure 6.5: Schematic of 1 V Bandgap reference generator circuit



Design Variables		
	Name	Value
1	VDD	3.3
2	R1	6.1359K

Figure 6.6: When $R1 = 6.1359k$ $V_{ref}=1V$.

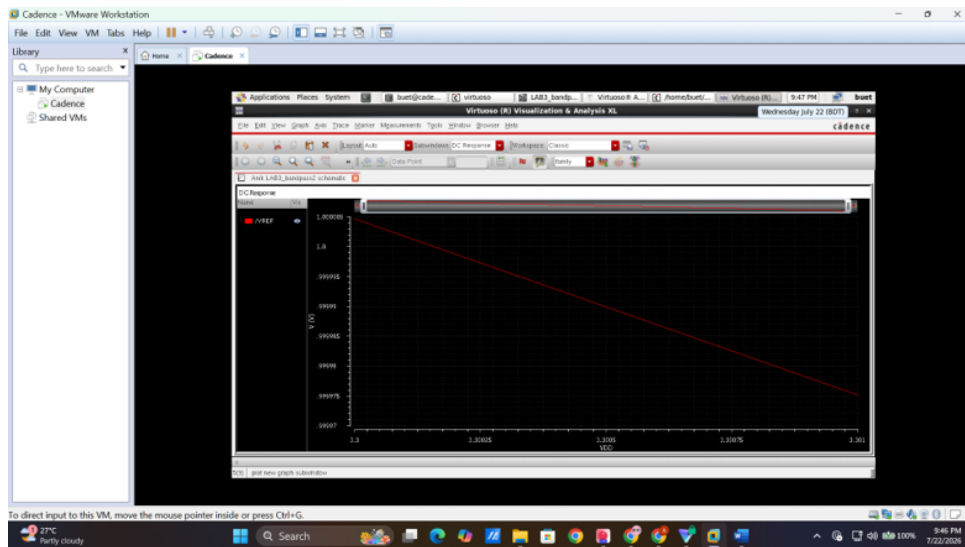


Figure 6.7: Graph of V_{ref}

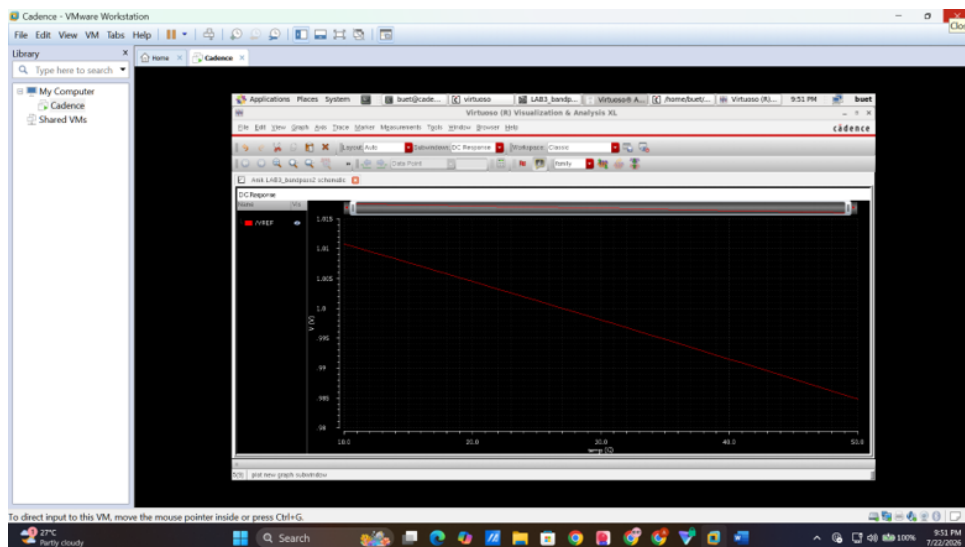


Figure 6.8: Temperature Sweep

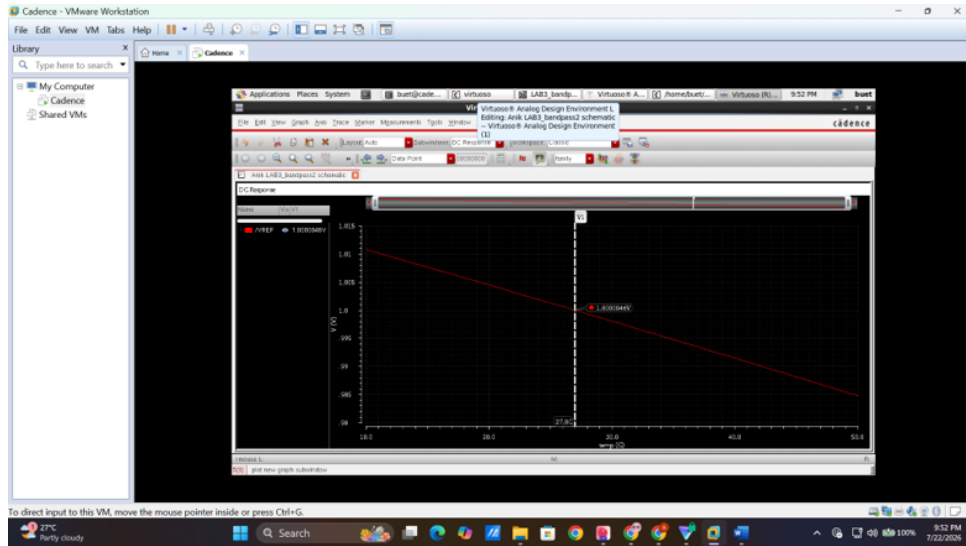


Figure 6.9: At 27 C, obtaining Tc value

$$TC = -644.337ppm/^{\circ}C \quad (6.5)$$

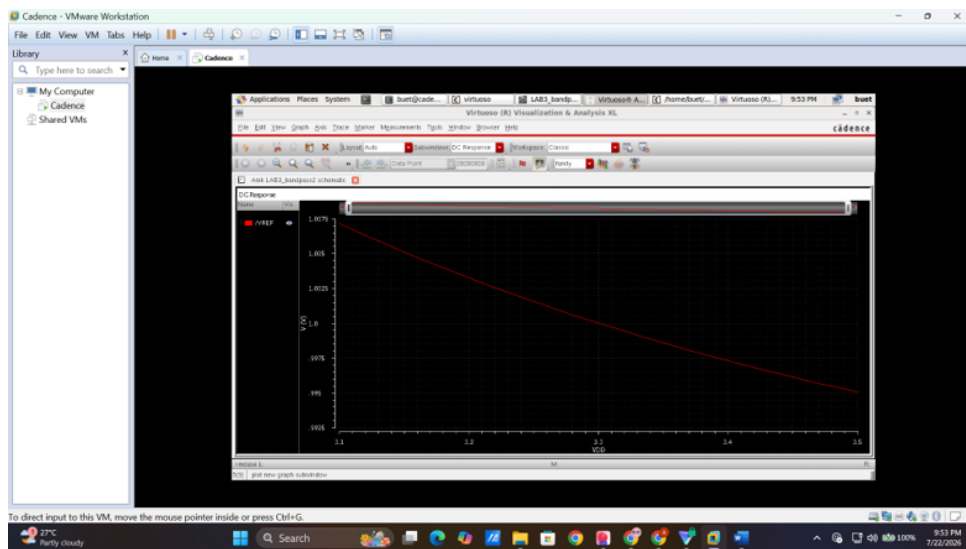


Figure 6.10: Variation of reference voltage with supply voltage

Chapter 7

CONCLUSION

In this experiment, a CMOS-based bandgap reference voltage generator was designed and simulated to study the generation of a stable reference voltage. The main objectives of the experiment, including circuit design, simulation, and performance analysis, were successfully achieved.

The simulation results showed that the designed circuit produced a reference voltage of $0.8136V$ with a supply voltage of $3.3V$, which is very close to the expected value of $0.814V$. This verifies the accuracy of the designed bandgap reference circuit and demonstrates the relationship between the theoretical concept and simulation results.

The temperature sweep analysis from $10^{\circ}C$ to $50^{\circ}C$ showed that the reference voltage changes slightly with temperature variation. The calculated temperature coefficient at room temperature ($27^{\circ}C$) was $-1.2066mV/^{\circ}C$, indicating the temperature stability of the circuit. The results confirm that the negative temperature coefficient of V_{BE} and the positive temperature coefficient of ΔV_{BE} are combined to reduce temperature dependency.

The supply voltage variation analysis from $3.1V$ to $3.5V$ showed that the reference voltage remains relatively stable, demonstrating good supply rejection capability of the bandgap reference circuit. These observations confirm the effectiveness of the temperature compensation technique used in bandgap reference design.

Although the designed circuit provides a stable reference voltage, further improvements can be achieved by optimizing transistor sizing, resistor matching, and operational amplifier characteristics. Advanced compensation techniques and precise device models can also be used to achieve a lower temperature coefficient and improved accuracy for high-performance analog applications.

Chapter 8

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